

Pandas Missing Values Extra Practice Questions

✓ DATASET

```
data = {  
    "Name": ["Amit", "Riya", None, "Pooja", "Karan", None],  
    "Age": [25, None, 30, 22, None, 28],  
    "Math": [78, 85, None, 90, 88, None],  
    "Science": [80, None, 75, None, 92, 85],  
    "English": [None, 70, 68, None, 75, 82],  
    "Salary": [50000, None, 60000, 55000, None, None]  
}
```

This dataset has:

- Missing names
- Missing ages
- Missing marks
- Missing salary

Task 1 — Show Missing Values

Print:

```
df.isnull()
```

Task 2 — Count Missing Values per Column

```
df.isnull().sum()
```

Task 3 — Count Total Missing Values

Count all missing values in entire DataFrame.

Task 4 — Remove All Rows with Missing Values

Use:

```
df.dropna()
```

Task 5 — Remove All Columns with Missing Values

Use:

```
df.dropna(axis=1)
```

Task 6 — Fill All Missing Values With 0

Use:

```
df.fillna(0)
```

Task 7 — Fill Age Column With Mean Age

```
df['Age'].fillna(df['Age'].mean())
```

Task 8 — Fill Salary With Median Salary

```
df['Salary'].fillna(df['Salary'].median())
```

Task 9 — Fill Missing Names With "Unknown"

```
df['Name'].fillna("Unknown")
```

Task 10 — Forward Fill the Dataset

```
df.fillna(method='ffill')
```

Task 11 — Backward Fill the Dataset

```
df.fillna(method='bfill')
```

Task 12 — Conditional Fill:

If English is missing → fill with **60**

Task 13 — Conditional Fill 2:

If Science is missing → fill with **subject average**
You must calculate mean first.

Task 14 — Fill Each Column Using Different Rules

Use a dictionary:

```
rules = {  
    "Age": df['Age'].median(),  
    "Name": "No Name",  
    "Salary": 0  
}
```

Task 15 — Simple Report

Print:

- how many missing values original dataset had
- how many missing values after filling
- number of rows before vs after dropna

(Just print numbers — no advanced logic needed)

